

General Directions: Complete all of the following questions. A perfect examination will earn 105 out of 100 points. Read the directions for each section carefully. Failure to comply with the directions may result in a loss of points. For example, writing two letters in one blank will receive 0.

Due Date: 11:59 p.m., Wednesday, December 18, 2024 on Canvas

- This is an open-book, open-notes, untimed examination.
- Do not to discuss this exam with anyone else (including bots), except for the instructor.
- If you are uncertain what a question is asking, you may contact the instructor (butlera@iastate.edu) for clarification; or you may write a brief explanation describing how you interpret the question.
- Save your completed exam in a standard document (.docx, .pdf, .rtf) or picture format. Or you may type your answers using Canvas's text submission box.
- Submit your exam on Canvas by the due date. Consult the syllabus for the late policy.

Note: The name(s) in parentheses refers to the reading(s) associated with the question.

Multiple Choice. Write the letter of the one best option in each blank. NOTE: The same letter may be used more than once; not every letter must be used. (3 points each; 105 points total)

- _____ 1. According to Leong and Selinger, what is implied by the widespread adoption of biometrics and multi-factor authentication by cybersecurity officers? (Leong and Selinger)
- A. No amount of employee training in cybersecurity practices will eliminate the vulnerability of employees to phishing, etc.
 - B. Dishonest anthropomorphism is sometimes not bad
 - C. Employees are lazy
 - D. All of the above
- _____ 2. Which of the following best illustrates a “responsibility gap”? (Mann, and others)
- A. A pharmacist fails to check the expiration date on the medicine before administering it
 - B. A human driver fails to brake before colliding with a self-driving car
 - C. A leashed dog on a walk with its owner bites the mailman
 - D. An ATM debits your account \$100 without dispensing any bills
- _____ 3. What asymmetry do Mann, and others identify in our assignment of praise and blame in responsibility gaps? (Mann, and others)
- A. When there are responsibility gaps, we tend to give credit but withhold blame
 - B. When there are responsibility gaps, we tend to assign blame but withhold credit
 - C. When there are responsibility gaps, we neither assign credit nor blame
 - D. When there are responsibility gaps, we both assign credit and blame
- _____ 4. Suppose that I use a chatbot to help write a paper. **Question:** According to Mann, and others, which use of the chatbot would make the chatbot a co-author? (Mann, and others)
- A. I write the entire paper and the chatbot provides the title for the paper
 - B. I write the entire paper and the chatbot corrects the spelling and grammar
 - C. The chatbot interprets the data, but I write the paper
 - D. None of the above

- _____ 5. What is Mann, and others' concern about "achievement gaps"? (Mann, and others)
- A. If humans working with AI are only blameworthy, then they never deserve credit for their accomplishments because they were assisted by AI
 - B. The existence of responsibility gaps implies that it is unclear or undetermined who deserves credit for accomplishments
 - C. People who work with AI are demoralized, so that they are unmotivated to aspire towards excellence
 - D. AI tools make users less skilled
- _____ 6. According to Nyholm, which of the following is required for "moral agency"? (Nyholm)
- A. The ability to recognize what has value
 - B. The ability to act for the sake of what is valuable
 - C. Both A and B
 - D. Neither A nor B
- _____ 7. Which of the following would show the success of machine ethics? (Nyholm)
- A. A machine operates without consideration of whether what it does is right or wrong
 - B. A machine operates in such a manner that it does the right thing because it recognizes that it is the right thing to do
 - C. A machine operates in such a manner that it does the right thing without any understanding of right or wrong
 - D. Machines succeed at world domination
- _____ 8. Suppose that morality cannot be reduced to a complete set of rules. **Question:** What response might be given by someone who defends machine ethics? (Nyholm)
- A. Moral agency requires judgment: moral agents select which rules to apply in different scenarios
 - B. Morality is context-sensitive: moral agents have the capacity to identify which rules apply in different scenarios
 - C. Morality does not consist in a set of rules, but in acting in ways that a virtuous agent would act, which an AI could acquire through training
 - D. None of the above
- _____ 9. Suppose that robots do not have or understand first-person experiences of suffering or desire (e.g., "I am suffering," etc.). **Question:** What would this show? (Nyholm, sect. 7.3.4)
- A. Robots cannot reason
 - B. Robots do not have emotions
 - C. Robots cannot correctly answer questions about right and wrong
 - D. Robots cannot act on moral reasons
- _____ 10. **True or False:** According to Nyholm, if a group is morally blameworthy, then each member of that group must be morally blameworthy. (Nyholm)
- A. True: group moral responsibility implies that each individual member is morally responsible
 - B. False: a group may be morally blameworthy even though nobody in the group is morally blameworthy
 - C. False: a group may be morally blameworthy, as long as at least one member of the group is morally blameworthy
 - D. Nyholm thinks group moral responsibility is nonsense

- _____ 11. If moral responsibility requires emotions, how can the harms produced by an emotionless robot be blameworthy, according to Nyholm? (Nyholm)
- A. The robot is a member of a team with at least one human member whose emotions guide the team's moral actions
 - B. The robot is sensitive to the dignity of human life
 - C. The robot is not morally blameworthy
- _____ 12. Which of the following harms would not be easily addressed by U.S. tort law? (Weaver)
- A. An autonomous robot surgeon nicks an artery, which requires canceling surgery
 - B. A human tries to repair an iPhone, which causes the iPhone to malfunction
 - C. A human supervisor fails to notice that the autonomous robotic sorting machine is crushing packages
 - D. A design flaw in an autonomous robot vacuum causes it to catch fire
- _____ 13. Which of the following reasons does Weaver offer against holding owners legally responsible for the harms caused by their robots? (Weaver)
- A. The fault lies with the manufacturer
 - B. Doing so would discourage people from owning robots
 - C. Doing so would be easy to enforce
 - D. Sometimes more than one person owns a robot
- _____ 14. Does Weaver's proposed way of handling liability claims for robot harms (on the model of Worker's Compensation) require attributing moral responsibility to robots? (Weaver)
- A. Yes: harms don't exist unless something is morally responsible for it
 - B. Yes: but Weaver does not explain what moral responsibility consists in
 - C. No: Weaver's solution avoids taking a stance on filling the responsibility gap
 - D. No: Weaver assigns moral blame to the injured person
- _____ 15. **True or False:** According to Asaro, autonomous weapons systems came into existence only recently, with the development of AI. (Asaro)
- A. True: autonomous weapons systems require computer technology
 - B. False: there are autonomous weapons systems that do not use AI
- _____ 16. Which of the following conditions on using autonomous weapons systems is not a requirement of international humanitarian law? (Asaro)
- A. The attack must not be much, much greater in scope than its justification
 - B. The attack must promote the greatest overall happiness
 - C. The weapons must not target innocent civilians
 - D. The attack must achieve a justified military objective
- _____ 17. Suppose that a precision system exists which (without the need for a human operator or supervisor) identifies and kills only enemy combatants. **Question:** Which ethical view would say that militaries are morally required to use this precision system? (Asaro)
- A. Utilitarianism
 - B. Deontology
 - C. International humanitarian law
 - D. None of the above

- _____ 18. How does Asaro propose to fill the responsibility gap created by the use of military robots? (Asaro)
- A. Robots supervising robots
 - B. Meaningful human control
 - C. Teams composed of humans and robots, each assigned separate tasks
 - D. Trick question: he does not think the responsibility gap can be filled
- _____ 19. According to Mittelstadt, what difficulty arises for AI ethics, because of the absence of a single professional society for AI? (Mittelstadt)
- A. No AI developers take seriously the impact of their design choices for users
 - B. No clear professional processes to hold AI developers responsible for harms
 - C. No opportunity to compare best practices with other companies
 - D. All of the above
- _____ 20. Which of the following would be a challenge to the personality-based moral justification of intellectual property? (Moore and Himma)
- A. The creator's personality changes
 - B. The patented process does not exhibit the patent-holder's personality
 - C. All of the above
 - D. None of the above
- _____ 21. Which of the following reasons does the rule utilitarian offer to justify intellectual property rights? (Moore and Himma)
- A. The personality of the creator is infused in their creation
 - B. Granting limited rights to one creator does not prevent granting limited rights to other creators
 - C. The best consequences result from conferring limited rights to creators so that they can profit from their creations
 - D. Respecting the dignity of humanity requires conferring limited rights to creators so that they can profit from their creations
- _____ 22. Why do Moore and Himma discuss the possibility of government-supported innovation contests? (Moore and Himma)
- A. To challenge the assumption that creators will be incentivized to create only if they can privately own their creations
 - B. To challenge the claim that government action is slow
 - C. To challenge the claim that intellectual products are "nonrivalrous"
 - D. To challenge the assumption that society benefits from public property
- _____ 23. How does a Lockean philosopher justify private property? (Moore and Himma)
- A. Because the creator's effort converts the public resource into something of value, the creator has rights of ownership
 - B. The creator's personality is infused into the creation
 - C. The owner is the first to call "dibs" on the item
 - D. The owner destroys the competition

- _____ 24. What is Locke's "proviso"? (Moore and Himma)
- A. The private property owner cannot destroy resources so that others cannot create
 - B. The private property owner cannot ruin the quality of resources so that others cannot create
 - C. The private property owner cannot hoard resources to make it impossible for others to create
 - D. All of the above
- _____ 25. What does it mean to say that information is "nonrivalrous"? (Moore and Himma)
- A. Truths are consistent with other truths, but not with falsehoods
 - B. Information wants to be free
 - C. Information cannot be treated as private property
 - D. One person's possessing some particular information does not prevent others from having it too
- _____ 26. **True or False:** According to Schur, sampling in music is always morally wrong. (Schur)
- A. True: sampling is always morally wrong
 - B. False: sampling may be conducted in a morally permissible way
- _____ 27. According to Schur, why is "authenticity" a core value for academic scholarship? (Schur)
- A. It gives the reader reason to trust the author
 - B. It assures the reader that the author has used the ideas of other authors accurately
 - C. It assures the reader that the work was produced by the author
 - D. All of the above
- _____ 28. **True or False:** Anyone who uses copyrighted work (including academic scholars and students) must always pay royalties or licensing fees. (Schur)
- A. True: any use of copyrighted work requires payment
 - B. False: some uses of copyrighted work fall under the "fair use" exception
- _____ 29. Suppose that authorship requires human creative choices. **Question:** According to Smits and Borghuis, would it automatically follow that no generative AI could ever be classified as authored? (Smits and Borghuis)
- A. Yes: there are no human decisions in generative AI
 - B. No: many generative AI outputs involve human decisions
- _____ 30. Suppose that no generative AI products are copyrighted. **Question:** According to Smits and Borghuis, would this deny creators the possibility of profiting from their AI creations? (Smits and Borghuis)
- A. Yes: all generative AI products would be forced into the public domain
 - B. Yes: creators could not deny anyone access to their creations
 - C. No: developers could still obtain copyrights or patents for their software
 - D. No: creators would profit from individually licensing their products
- _____ 31. According to Smits and Borghuis, do human artists directly compete with generative AI "qwrks" for customers? (Smits and Borghuis)
- A. Yes: and generative AI dominates the marketplace because of its scale and speed
 - B. Yes: and human artists dominate the marketplace because of customers' desire for authenticity
 - C. Not necessarily

- _____ 32. Which of the following reasons would the U.S. Copyright provide for rejecting copyright for a generative AI “qwrk”? (Smits and Borghuis)
- A. Generative AI is not human
 - B. Generative AI operates automatically
 - C. Generative AI produced by a mere mechanical process
 - D. All of the above
- _____ 33. Why do Bender, and others call chatbots “stochastic parrots”? (Bender, and others)
- A. Chatbots put sentences together based on frequencies of word pairings in their training data, not based on how the world is
 - B. Anthropomorphism: Bender, and others apply an animal form to machines
 - C. That is the name assigned to them by the first creators of chatbots
 - D. Parrots sometimes hallucinate
- _____ 34. Do Bender, and others think that making training datasets larger will address social biases? (Bender, and others)
- A. Yes: more data ensures greater accuracy, reducing bias
 - B. Yes: more data ensures that a greater variety of views are included, reducing bias
 - C. Yes: more data ensures that marginalized groups are included, reducing bias
 - D. No
- _____ 35. Why do Bender, and others think that it is important for AI companies to have documentation about their training datasets? (Bender, and others)
- A. Accountability
 - B. Transparency
 - C. To make it possible to audit
 - D. All of the above